

# Sri Lanka Institute of Information Technology



IT1040 - Fundamentals of Computing  
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Smart Home and Environment Monitoring Systems

Proposal Document

Group 01.35

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## **1. Background**

- The rise of smart technologies is transforming how we live and interact with our environment.
- Homes and buildings are expected to respond automatically to changing conditions.
- Our project explores the use of affordable, low-tech systems that provide smart automation.
- These systems address common local needs such as managing floods, controlling lights, and monitoring temperature.
- The focus is on creating practical solutions that are simple, cost-effective, and applicable in Sri Lankan contexts. From managing light and temperature to preventing floods, we believe technology should work for everyone, not just those with access to high end solutions.



## **2. Problem**

### **1. Wastage of Energy at Home**

Energy is often wasted in households due to lights being left on unnecessarily. This leads to higher electricity bills and increased environmental impact from avoidable energy use.

### **2. Discomfort from Environmental Conditions & Manual Control**

#### **Limitations**

Many homes suffer from excessive heat and sunlight, yet most systems still rely on manual control. This causes discomfort and inefficient energy usage, especially when cooling systems are overused.

### **3. Impact of Sudden Utility Failures**

Unexpected failures in essential utilities, such as water supply, can occur without warning. These disruptions cause immediate inconvenience and leave no time for preparation.

### **4. Localized Consequences of Natural Hazards**

Flood-prone areas like Ratnapura frequently face severe flooding due to the overflow of the Kalu Ganga River. Floodwaters damage homes, destroy crops, and displace residents often without early warning.

## **Motivation**

- **Optimizing Energy Use:** There is a clear need for smart lighting systems that operate automatically based on occupancy and ambient light conditions.
- **Enhancing comfort in homes** requires automated solutions that adjust fans and window blinds without manual input. These systems can respond to temperature and sunlight changes, improving energy efficiency.
- **Proactive Solutions for Utility Reliability:** Our automated water switching system can detect service interruptions (water interruptions) and activate backups, minimizing inconvenience during emergencies.
- **Opportunity for Safer Flood Preparedness:** With early warning systems, communities like Ratnapura can prepare in advance, protect their homes, and stay safe during sudden floods.

### 3. Aim and objectives

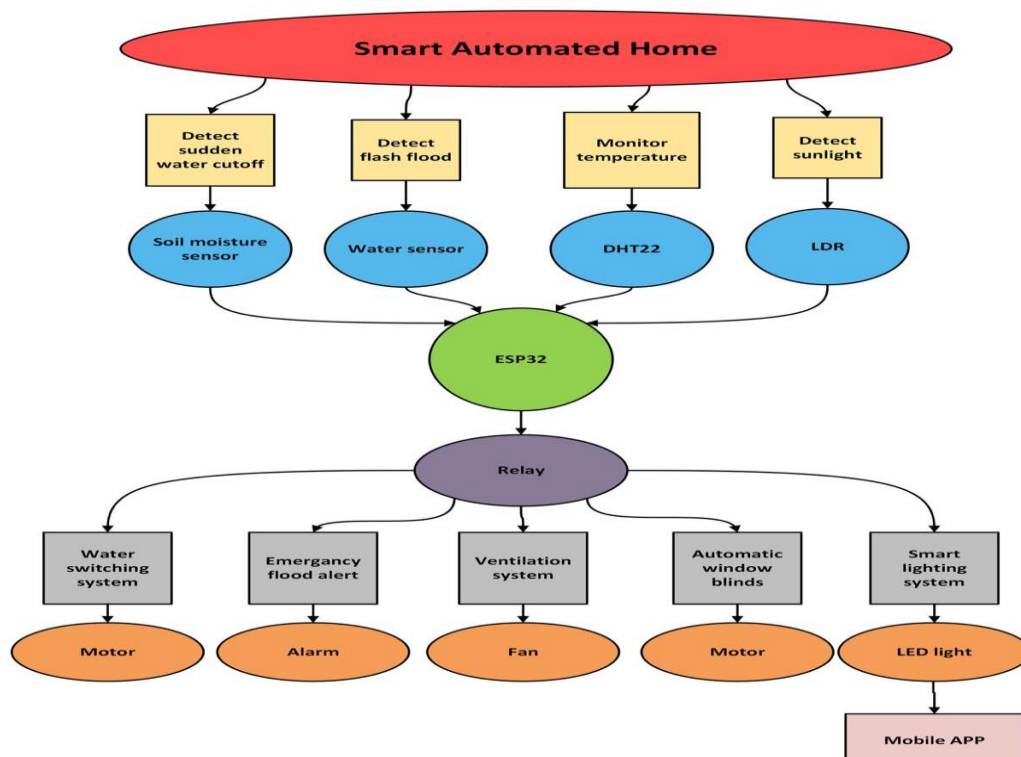
#### Aim

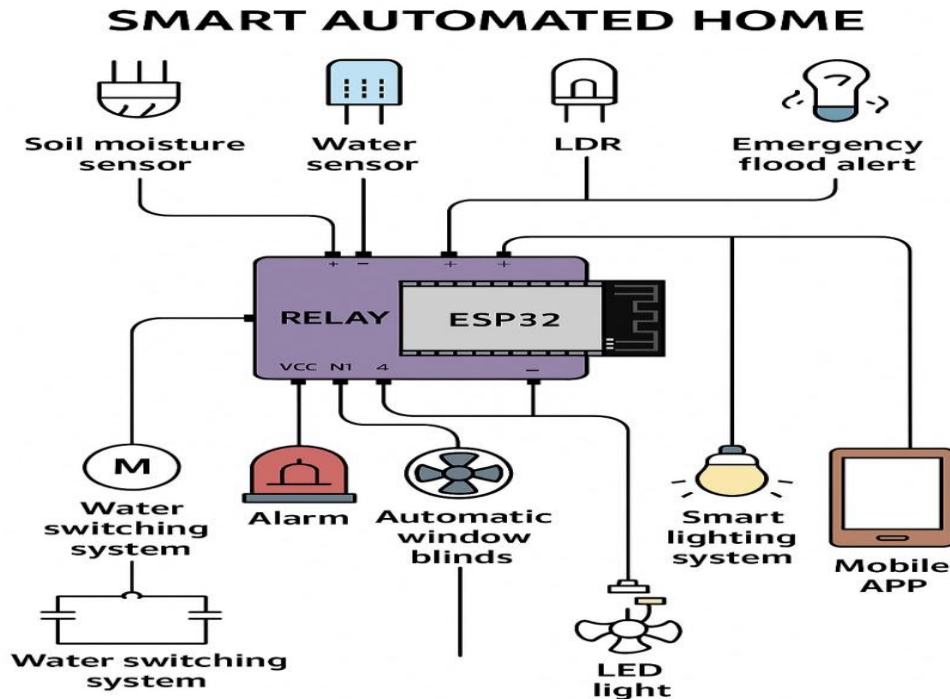
Our aim is to design a smart home automation system that enhances comfort, efficiency, and safety through intelligent control of responsive and adaptive technologies.

#### Objectives:

- Optimize natural light usage and indoor temperature regulation.
- Maintain indoor thermal comfort with minimal energy use.
- Reduce energy wastage from unnecessary lighting.
- Alert occupants immediately during supply interruptions to prevent inconvenience or damage.
- Improve household preparedness and automatic system responses (e.g., closing windows during storms or triggering emergency lighting).

### 4. System Diagram





## **5. Methodology**

### **1. System Design and Planning**

The project will begin with a complete design phase, during which we will outline the smart automated housing system architecture and performance. This includes creating the housing system layout and determining the placements for sensors and other configurations.

### **2. Tools and Technologies**

**ESP 32 :-**The central control unit that manages all the devices and sensors.

**Actuators:-** Motors, Lights, Fan, Automatic window blinders

**Sensors: -** LDR Sensor, DHT22, Water Sensor, Ultrasonic Sensor (HC-SR04)

**Power Supply**

**Bread Board**

**Broadband (Jumper Wires)**

#### **- Automated Window Blinds Implementation**

Hardware: Light Sensor (LDR with 10k $\Omega$  resistor), Servo Motor (SG90)

Software: Logic was programmed to trigger motor actions based on light threshold values.

#### **-Temperature Responsive Fan System**

Hardware: MLX90614 Infrared Temperature Sensor, OLED Display (I2C interface)

Software: Fans were programmed to turn on/off based on preset temperature ranges.

#### **-Smart Lighting System**

Hardware: PIR Motion Sensor, Light Sensor (LDR), LED Bulb or 5V LED Strip

Software: Lights can manually turn on through a mobile application

#### **-Sudden Utility Cutoff (Water Cutoff) Detection & Automated Switching**

Hardware: Water Level Sensor / Float Switch, Submersible Pump, Relay Module

Software: Submersible pump was programmed to turn on/off based on water flow.

#### **-Natural Hazard Monitoring (Flash Flood Detection)**

Hardware: Ultrasonic Sensor (HC-SR04), Rain Sensor Module, Buzzer, LED Indicator

Software: Buzzer was programmed to automatically activate when a potential flash flood is detected.

### **3.Integration and Testing**

All subsystems were connected to a central control unit using ESP32.

Developed a simple mobile interface(eg:Blynk) for system monitoring and manual overrides.

Thoroughly tested individual modules and the integrated system under various environmental conditions.

### **6.Evaluation Method**

To ensure our systems are effective and reliable, we will use a comprehensive evaluation method that covers both technical and practical aspects. Each system will be tested using the following criteria:

- 1. Functionality Testing:** Verify that each system performs its intended task under normal conditions (e.g., the flood alert activates when water reaches a set level).
- 2. Accuracy and Responsiveness:** Compare sensor outputs against known benchmarks and measure the time it takes for the system to respond to changes in the environment.

- 3. Reliability (Stability Over Time):** Run each system over extended periods and in repeated cycles to confirm it behaves consistently without failure
- 4. Power Efficiency:** Measure power consumption of each system using multimeters or power monitors. Ensure the ESP32 and components minimize energy use in idle states.
- 5. User-Friendliness:** Evaluate how easy it is to install and use each system. Feedback from test users will help identify design improvements.

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